MIL-C-23961A(SHIPS)
23 December 1965
SUPERSEDING
MIL-C-23961(SHIPS)
31 January 1964
(See 6.5)

MILITARY SPECIFICATION

COMPRESSORS, HIGH PRESSURE AIR, 100 C.F.M., NAVAL SHIPBOARD SERVICE

1. SCOPE

1.1 This specification covers electric motor driven high pressure air compressors (5500 pounds per square inch gage (p.s.i.g.), maximum) for installation aboard Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
QQ-S-763 - Steel Bars, Shapes, and Forgings - Corrosion Resisting.
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.
GGG-P-781 - Puller and Puller Kit, Mechanical, and Mechanical Puller Attachment.

MILITARY

MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-B-857 - Bolts, Nuts and Studs.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-F-1183 - Fittings, Tube Cast, Bronze, Silver Brazing.
MIL-C-2212 - Controllers, Alternating Current, Naval Shipboard.
MIL-L-9000 - Lubricating Oil, Internal Combustion Engine, Diesel.
MIL-P-15024 - Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).
MIL-T-15377 - Temperature Monitor Equipment, Naval Shipboard.
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard, Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-M-17060 - Motors, 60-Cycle, Alternating-Current, Integral Horsepower (Shipboard Use).
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories and Instruments); Packaging of.
MIL-L-17331 - Lubricating Oil, Steam Turbine (Non-corrosive).
MIL-P-17545 - Primer Coating, Steel Ship Maintenance, Alkyd - Red Lead Type, Formula No. 116.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-B-17931 - Bearings, Ball, Annular, for Quiet Operation.
MIL-A-18001 - Anodes, Corrosion Preventive, Zinc, Slab Disc and Rod Shaped.
MIL-G-18997 - Gages, Pressure, Dial Indicating, Bourdon Tube.

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MILITARY (Cont'd)
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-V-20065 - Valve, Relief, Naval Shipboard.
MIL-F-21467 - Fittings, Flareless, Fluid Connection (Shipboard Use).
MIL-V-22549 - Valves, Angle, Relief, Naval Shipboard for Gas Service (Sizes 2-Inches IPS and Below).
MIL-L-26087 - Lubricating Oil, Reciprocating Compressor, Ground Support.

STANDARD

MILITARY
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.
MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships.

DRAWINGS

BUREAU OF SHIPS
810-1385850 - Piping, Gage, for All Service.
810-1385884 - Unions, Fittings and Adapters, Butt and Socket Welding, 6000 PSI, WOG and OXY (IPS).
810-1385941 - Fittings, Silver Brazing, WOG, 3000 PSI for U/T Inspection.
810-1385942 - Fittings, Bronze Silver Brazing, WOG, for U/T Inspection.
810-1385943 - Unions, Silver Brazing, 3000 PSI, WOG, IPS for U/T Inspection.
810-1385944 - Fittings, Silver Brazing, OD, 3000 PSI, WOG for U/T Inspection.
810-1385945 - Bushings, Reducing, Copper-Nickel, Silver Brazing, WOG for U/T Inspection.
810-1385946 - Unions, Bronze Silver Brazing, WOG for U/T Inspection.
810-1385948 - Unions, Silver Brazing, OD, 3000 PSI, WOG for U/T Inspection.
810-1385950 - Bosses, Non-Ferrous, Piping for U/T Inspection.
810-1385951 - Bushings, Reducing, Copper-Nickel Silver Brazing, IPS, 3000 PSI, WOG for U/T Inspection.
810-1385952 - Bushings, Reducing, Copper-Nickel Silver Brazing, OD, 3000 PSI, WOG for U/T Inspection.

PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 250-660-30 - A Guide for Design of Shock Resistant Naval Equipment.
NAVSHIPS 0900 - 001-7000 (NAVSHIPS 250-634-6) - Fabrication and Inspection of Brazed Piping Systems.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

COMPRESSED AIR AND GAS INSTITUTE
ASME Power Test Code (PTC9-1954-Compressed Air Handbook, 3rd edition).

(Application for copies should be addressed to the Compressed Air and Gas Institute, 90 West Street, New York, New York.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C.)

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, Park Avenue at 33rd Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Capacity.-

3.2.1 The nominal capacity of the compressors shall be 100 cubic feet per minute (c.f.m.) of standard air. The rated capacity of the compressors shall be 30 cubic feet of compressed air per hour (standard atmospheric suction conditions) at a discharge temperature of 125° Fahrenheit (F.) and a discharge pressure of 3000 p.s.i.g., and shall deliver at least 18 cubic feet of compressed air per hour at 5000 p.s.i.g. The rated capacity shall be obtained at a cooling water inspection temperature of 85°F.

3.2.1.1 Standard atmospheric suction conditions, or standard air is defined as air at 14.7 pounds per square inch absolute (p.s.i.a.) pressure, 68°F. temperature, 36 percent relative humidity (density 0.0750 pounds per cubic foot (p.c.f.).

3.2.2 Means shall be provided for manually adjusting the compressor capacity to any value from 27 cubic feet per hour (c.f.h.) or below to 30 c.f.h. or above, measured at 3000 p.s.i.g. discharge pressure.

3.3 Description.-

3.3.1 Each compressor shall be multistage, reciprocating, crosshead type, vertical or horizontal design, with an intercooler and moisture separator after each stage of compression.

3.3.2 The compressor may be either direct driven or vee-belt driven. If vee-belt drive is provided, static free belts and protective guards shall also be provided. Vee-belt design shall be based on 125 percent rated torque capacity.

3.3.3 A complete compressor unit shall consist of the compressor and motor mounted on a common bedplate and shall include all necessary appurtenances except that motor controller, pressure switch and suction silencer may be separately mounted. Safety features, including guards for all exposed moving parts, shall be incorporated in the design.

3.3.4 The dry weight of compressor with motor, less motor controller, but including all other appurtenances, shall not exceed 7500 pounds. Compressor, including motor, shall not exceed the following dimensions including space for disassembly, but exclusive of operating space:

(a) Length - 7 feet 8 inches
(b) Width - 4 feet
(c) Height - 6 feet 2 inches

3.3.5 The compressor shall operate satisfactorily when subjected to the following ship motions regardless of whether the bedplate is installed in the fore-and-aft or athwartships position:

(a) Up to 15 degrees in any direction (permanently inclined).
(b) With the ship rolling, up to 30 degrees from the vertical to either side (with direction of crankshaft in either the athwartships or fore-and-aft position).
(c) With the ship pitching, 10 degrees up and down from the normal horizontal plane.

3.4 Reliability assurance program.-

3.4.1 The principle of maximum reliability is paramount, and no compromise of this principle shall be made with any other basic requirements of design. Where wear or corrosion is unavoidable, the parts subjected to such wear or corrosion shall be of the best material for the purpose in order to reduce those

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detrimental effects to a minimum. It is the intention of this specification to obtain air compressors of such design that they will have an operating life of at least 50,000 hours with no limitation on the number of starts and stops, and without major overhauls of less than 10,000 hour intervals. Replacement of easily replaceable wearing parts such as cylinder suction and discharge air valves should not be required at intervals of less than 1000 hours. Replacement of piston rings should not be required at intervals of less than 1000 hours for the high pressure stages and of not less than 2000 hours for the first and second stages.

3.4.1.1 The designer should take cognizance of the conditions under which equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. For example, parts should be so designed that it is impossible to replace them in a wrong position; and adjacent studs should be of the same length, or of large difference in lengths, so that a mistake will not easily occur; serrated surfaces should not be used for accurate positioning unless specifically approved by the Bureau of Ships since areas where strong wrenching would cause damage should be avoided.

3.4.1.2 The contractor shall establish and maintain an effective reliability assurance program including the following:

(a) Design reviews - The reliability assurance program shall include provisions for the reliability review and evaluation of designs as an integral part of the contractor's engineering design procedures. Design or engineering changes occurring during development or production shall be subjected to comparable review procedures.
(b) Production control and monitoring - The reliability assurance program shall provide an economical and effective system of production control and monitoring to assure that reliability achieved in design is maintained during production.
(c) Subcontractor and vendor reliability - The reliability assurance program shall include provisions to assure subcontractor and vendor selection and performance consistent with the reliability requirements of the contract and applicable portions of this specification.
(d) Failure reporting, analysis and feedback - The reliability assurance program shall incorporate a formalized system for recording, collecting, and analyzing all failures that occur during all testing, installation and operation through the tenure of the contract. Analysis shall be fed back to contractor's engineering, management, and production activities on a timely basis. Failure reports received from using activity shall be integrated into this program for trouble analysis and for experience considerations for future problem solving and design improvement.

3.4.2 Each compressor shall be capable of continuously supplying the specified quantity of air at the normal discharge pressures and standard design conditions (see 3.4.4). All compressors shall be capable of satisfactory operation at an overload pressure of 500 pounds per square inch (p.s.i.) over their rated discharge pressure for a period of 2 hours in any 4-hour period.

3.4.3 The temperature of air discharges from any stage of compression shall not exceed 300°F. for final air pressures up through 3000 p.s.i.g., and 340°F. for pressures above 3000 p.s.i.g. but not exceeding 5000 p.s.i.g. The temperature shall be measured by temperature sensing devices (see 3.4.21.3.1) located in the air stream as close as physically practicable to the discharge side of the discharge valves.

3.4.4 The compressor, complete with its motor drive and all appurtenances, shall be designed for and shall operate satisfactorily under the following range of conditions encountered onboard ship:

(a) Ambient and suction temperature - 40° to 122°F.
(b) Cooling water injection temperature - 28° to 85°F.

3.4.5 Shockproofness. -

3.4.5.1 The design of all complete air compressor units shall be such that they are capable of passing the type A, H.I. (high-impact) shock tests specified in MIL-S-901. Grade A, class III equipment in accordance with MIL-S-901 is required. A shock test with the compressor solidly mounted will be considered proof of class III capability (see 4.5.4). NAVSHIPS 250-660-30 may be used as a guide for design of shock resistant Naval equipment.

3.4.5.2 Shockmounts shall not be used unless bureau or agency approval of the mount and its proposed application is given.

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3.4.5.3 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be no more than 1/32 inch larger than the bolts up to 3/4 inch size, and no more than 1/16 inch larger than bolts of greater than 3/4 inch size. Mounting bolts for fastening the equipment shall conform to grade 2 of MIL-B-857.

3.4.6 Noise and vibration. - Each complete compressor unit, including the motor, shall be designed to obtain the quietest possible operation consistent with compressor reliability and accessibility for maintenance and repair.

3.4.7 The crankcase, frame, housing, cylinders, and other fixed parts or attached component parts of the assembly shall be rigidly secured and held in alignment by dowel pins, fitted bolts, keys, rabbeted or tongue and groove joints, or other suitable means.

3.4.8 Air cylinders. - Air cylinders shall be provided with ample water jackets. So far as practicable, handholes shall be provided for access to the jacket spaces. Cylinder liners shall be provided and shall be held rigidly in place. All shoulders where piston rings may catch shall be avoided and liners shall be properly beveled. Back off, jacking screws or other suitable means shall be provided to assist in disassembly at all gasketed joints between air cylinders and between air cylinders and cylinder heads.

3.4.9 Pistons. - Piston speeds shall not exceed 720 feet per minute (f.p.m.). In addition, nominal rotative speed of the crankshaft shall not exceed 720 revolutions per minute (r.p.m.) unless otherwise approved by the Bureau of Ships. Step seal design piston rings below 4 inches nominal diameter shall be of a ductile material. When iron rings are used with aluminum pistons, the aluminum shall be hardened by anodyzing to prevent excessive ring groove wear.

3.4.10 Connecting rods. - The crank end of connecting rods shall be arranged for renewal of crankpin bearings and for such adjustment for bearing wear as may be necessary for the proper operation and maintenance of the compressor. Bolts and nuts for securing crankpin bearings shall be positively locked. Any takeup bolts and nuts shall be positively locked. Upper end of connecting rods shall be of one-piece construction.

3.4.11 Crankshaft. - The crankshaft shall be a single forging or casting. The cranks shall be arranged to give the best possible running balance. Counterweights shall be fitted if necessary. The design shall be such as to permit convenient use of a portable tachometer.

3.4.12 Flywheels. - Each compressor shall be equipped with a flywheel of sufficient size to provide smooth running of the compressor and motor. Flywheels shall be keyed to the crankshaft and shall be pressed or securely clamped thereon. A separate flywheel will not be required if ample flywheel effect is otherwise provided.

3.4.13 Bearings. -

3.4.13.1 Main bearings shall be replaceable and may be of a rolling contact type in accordance with MIL-S-17331 or FF-B-185 or a sliding contact type, as approved by the Bureau of Ships.

3.4.13.2 Crank pin bearings shall be precision type replaceable bearings. All connecting rod bearings and bushings shall be renewable.

3.4.14 Lubrication. -

3.4.14.1 Lubricating oil for cylinders shall be in accordance with MIL-L-26087 except that it need not comply with the viscosity specified therein. Crankcase oil shall be in accordance with MIL-L-26087 or MIL-L-17731 or MIL-L-9000.

3.4.14.2 An adequate and self-contained lubricating system of the forced feed type shall be provided for the crankpin bearings, gears and pinions and for the main bearings if not antifriction type. A suitable oil sump or reservoir shall be provided, and the oil shall be drawn from the reservoir by one or more pumps which are driven from the compressor crankshaft or motor shaft. The arrangement of reservoir and pumps shall be such that oil will not be lost from the reservoir nor the pump lose suction due to roll or pitch of the ship. The reservoir shall be provided with suitable fittings for filling and draining. The drain fitting shall be arranged so that the reservoir can be completely drained when the compressor base is level.

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An oil level gage, marked to show the high, low, and normal oil levels, shall be included. Gage glasses, if used, shall be renewable without draining the oil from the reservoir.

3.4.14.3 A full flow oil filter, preferably of a self-cleaning type having an overpressure bypass, shall be installed in the pump discharge line to filter out solid particles which might damage the compressor. The filters shall not pass particles larger than 25 microns. Suitable means shall be provided for regulating the pressure of the lubricant. A device actuated by oil system pressure shall be provided to shut the compressor down in event the oil pressure is not sufficient for proper compressor operation.

3.4.14.4 The temperature of lubricant in the crankcase shall not exceed 180°F. Oil coolers shall be provided if necessary to prevent overtemperature.

3.4.14.5 Air cylinders shall be lubricated by means of a mechanical forced feed lubricator mechanically driven from the compressor or by a separate electric motor. If driven by a separate motor, the motor shall be electrically interlocked with the compressor motor to prevent starting the compressor without oil in the cylinders. The lubricator shall be provided with its own oil reservoir of sufficient capacity to supply cylinder lubrication for at least 8 hours of continuous operation at the manufacturer's specified normal delivery rate of the lubricator, but in no case of less than 5 pints capacity above the indicated low level. The cylinder lubrication system shall include the following:

(a) Means for adjusting the delivery rate of each individual cylinder over a wide range above and below the normal operating flow rate.
The recommended maximum flow rates and range of flow rates shall be indicated on the drawings.
(b) A handwheel or crank for operating the lubricator manually.
(c) Visual and quantitative flow indication to each cylinder.
(d) A check valve in each delivery line at the point where the line connects to the air system.
(e) A check valve at each lubricator discharge connection if necessary to prevent the lubricant from draining back to the lubricator reservoir after compressor shutdown.
(f) Means for checking oil level in lubricator reservoir.

3.4.15 Bedplates. - The bedplates shall be continuous under the compressor and its driving unit. They shall be machined to receive the intermediate frames or housing, or may be integral therewith. All top and bottom bearing surfaces shall be machine finished. Bedplates shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship without disturbing the alignment of the assembled units; and such that the normal distortion, weaving or vibration of the supporting structures onboard ship cannot cause misalignment of compressors.

3.4.16 Housing. - The housing shall consist of cast frames, cast frames and forged steel tie rods, or a structural steel welded assembly which shall support the cylinders. The crankpits and the housing shall be wholly enclosed to prevent the splashing of oil. Access openings with covers shall be provided for observation and adjustment of the bearings, and where necessary, breather openings fitted with screens shall be provided. The crankcase vent or breather shall be located as far from the compressor intake as possible.

3.4.17 Air valves. - The air valves shall be of an efficient and durable design and shall be readily accessible for inspection and overhaul.

3.4.18 Cooling system. -

3.4.18.1 General design. - The compressor shall be provided with an efficient water cooling system, including water jacketing for cylinders, intercoolers and aftercooler. The temperature of air delivered from the aftercooler shall not exceed 125°F. under rated operating conditions.

3.4.18.1.1 At any gasketed joint in the cooling water system where failure or leakage of a gasket or "O" ring would permit the entry of coolant into the air stream or of compressed air into the cooling system, a double seal, vented to atmosphere between the air seal and the coolant seal, shall be provided.

3.4.18.1.2 The cooling water outlet shall be provided with a water flow indicator of the "look-through" type, having an element which moves with flow, to permit observation of water flow and the detection of air in the cooling system. The cooling water outlet shall also be provided with a globe type manually operated valve for controlling the rate of flow through the cooling system.

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3. 4. 18. 1. 3 Provision shall be made for automatically venting entrained air from pockets or high points in the cooling water jackets of cylinders, cylinder heads and coolers. Provision shall also be made for completely draining all parts of the circulating water system.

3. 4. 18. 2 All parts of the cooling system in contact with water shall be of corrosion resisting material and shall be designed for the use of salt (sea) water supplied from the ship's flushing or fire mains at a maximum working pressure of 175 p. s. i. g. and a minimum working pressure of 50 p. s. i. g. The cooling system shall include an attached valve for positive and automatic control of the cooling water so that the water flow is maintained while the compressor is running and is stopped when the compressor is not running. The water flow should stop within 10 seconds after the compressor is stopped. A solenoid type valve shall not be used.

3. 4. 18. 2. 1 The cooling system shall be protected from electrolytic action by the insertion of replaceable zinc anodes designed to MIL-A-18001. The zinc anodes shall have as large a surface area as is practicable for maximum electrolytic protection without causing excessive turbulance. The zinc anodes shall be easily removable for inspection and replacement.

3. 4. 18. 3 The Bureau of Ships will give consideration to a closed fresh water cooling system employing liquids other than sea water, provided it can be demonstrated that there will be a saving thereby in weight or cost, or other definite advantages. In any case the ultimate heat sink will be sea water and the heat exchanger shall be furnished integral with the compressor. If an alternate design is proposed, then all departures from specification requirements shall be thoroughly described giving reasons and advantages to be expected. The closed system shall include a circulating pump, surge tank, water flow indicator and an effective rust inhibitor. If the pump is separately motor driven, the pump motor shall be electrically interlocked with the compressor motor.

3. 4. 18. 4 Lubricating oil coolers furnished as separate accessories shall conform to MIL-C-15730. Such coolers fabricated as an integral part of the compressor shall conform to the basic requirements of MIL-C-15730.

3. 4. 18. 5 Air coolers. - Intercoolers and aftercoolers shall be of the straight-tube type or of a type having straight tubes with a single return bend. For air pressures of 250 p. s. i. g. or higher, air shall flow through the cooler tubes and water flow over and around the tubes. For air pressures below 250 p. s. i. g., air may flow either through or around the tubes at the manufacturer's option. Cooler tubes shall be rolled, welded or brazed into tube sheets or headers, except that rolled tubes shall not be used for pressures above 250 p. s. i. g. Provision shall be made for prevention of damage due to the expansion of tube nests. Stay rods as necessary shall be provided for support of tube sheets and air heads. Each cooler tube nest shall be a self-contained unit so that it may be easily removed from the compressor as a unit for inspection and cleaning. Suitable air relief valves shall be fitted on or after each air cooler. Suitable water relief valves shall be fitted at the high point of the water side of each air cooler except for first and second stage coolers. Water relief valves shall be set at 185 p. s. i. (see 3. 4. 23).

3. 4. 19 Condensate drainage system. -

3. 4. 19. 1 Efficient moisture separators shall be provided after each intercooler and aftercooler for removing free moisture and oil from the air. Each separator shall be sized to require drainage not more often than every 30 minutes under the maximum condition of 50 percent relative humidity and 122°F. temperature of intake air, and the minimum condition of 28°F. cooling water temperatures in the range of 28°F. through 85°F.

3. 4. 19. 2 Each separator shall have a drain connection at its low point leading to an automatic separator drain device mechanically or electrically timed to completely drain each separator in sequence from first to final stage not less frequently than every 30 minutes. Drainage from the automatic drain device shall be led to an attached condensate accumulator to be stored under pressure until all stages have been drained, after which the automatic drain device shall permit the pressurized accumulator to discharge to ambient pressure through a drain connection on the compressor. This drain connection shall be suitable for attachment to ship's piping for drainage disposal. Means shall be incorporated for automatic discharge of the condensate accumulator each time the compressor stops in order to avoid full or partial pressurization of the drain system during shutdown.

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3.4.19.3 Each drain line between the separator and the automatic valve shall be provided with a suitable sight flow indicator, a check valve to prevent backflow during possible malfunction, and a manually operated valve to bypass the automatic device, thus permitting manual drainage direct to the drain connection. The sight flow indicators and the manual bypass valves shall be located at the designated operating side of the compressor in such a manner as to be readily observed and manipulated by the operator.

3.4.19.4 The condensate accumulator shall be connected to a separate gage on the gage board. This gage shall have a maximum scale reading no higher than the maximum operating pressure (or the next available scale reading above) so significant pressure indication can be obtained at the lowest stage pressure.

3.4.19.5 The quantity of air lost as the result of operation of the automatic drain system shall be kept to a minimum. This shall not reduce the required output of the compressors as required by 3.2.1.

3.4.19.6 At the manufacturer's option, the separator drain system may be arranged to provide automatic drainage of only the interstage separators provided the final separator is sized so that it will require drainage not more often than every 24 hours of compressor operation under the conditions described in 3.4.19.1.

3.4.19.7 The Bureau of Ships will give consideration to an automatic condensate drainage system differing from that described in 3.4.19.2 to 3.4.19.4 inclusive, provided it is demonstrated by actual test that the same end result is accomplished.

3.4.20 Unloading. - Each compressor shall be fitted with automatic unloading devices. These devices shall discharge residual air pressure to atmosphere immediately after the compressor is stopped, so that the compressor will always be unloaded when starting.

3.4.21 Fittings. -

3.4.21.1 Pressure gages. - All pressure gages shall be 3-1/2 inch dial size safety type gages in accordance with MIL-G-18997 and shall be mounted on a common gage board. The gage board and its mounting brackets shall be designed so that the board may be relocated to a bulkhead. The brackets shall be readily removable from the board.

3.4.21.1.1 Pressure gages shall be provided to indicate the air-delivery pressure from each stage of compression, the pressure in the circulating water system, and the lubricating oil pressure. All gages on the gage board shall be suitably identified by the adjacent attachment of identification plates on the gage board. Each gage shall be provided with means to indicate the normal desired working pressure, and such working pressures shall be indicated on drawings and in manuals.

3.4.21.2 Gage piping. - Gage piping shall be in accordance with Drawing 810-1385850, except that flared tubing fittings and tapered pipe threads will not be permitted, and except that root valves are not required for any gage line connected to a system which will be automatically depressurized when the compressor stops. Gage cocks may have test plugs in the valve body or stem in lieu of separate test plug fittings.

3.4.21.3 Temperature instrument. - A temperature indicating and alarm instrument similar to that specified in MIL-T-15377 shall be provided on each air compressor. A selector switch shall be provided to read at least twelve temperatures, and suitable circuits to automatically shutdown the compressor in the event any one of the selected temperature sensitive points (as many as five may be selected) exceeds a predetermined amount. The shutdown temperatures shall be individually adjustable over a range of 200°F. to 500°F. An alarm indicating light shall be mounted on the face of the instrument which will operate in conjunction with the shutdown feature. The instrument shall be located with its instrument panel readily accessible on the operating side of the compressor.

3.4.21.3.1 Temperature sensing elements shall be provided to detect the inlet and outlet temperatures at each stage of compression, the final air discharge temperature, the lubricating oil temperature in the reservoir, the cooling water outlet temperature, and the temperature at any other points considered by the manufacturer to be critical.

3.4.21.3.2 The sensing elements shall be inserted directly into the air, water or oil without the use of separate wells. Elements inserted in air streams shall be capable of withstanding 5500 p.s.i.g.

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pressure. Elements inserted in water shall be corrosion resistant. The lead wires from the sensing elements to the indicating and alarm instrument case shall be two copper wires incased in suitable Navy type cable. Lead wires shall be readily disconnected from the sensing elements without removal of the element from the compressor and may be grouped into a harness or bank for entry into the instrument case through a multiple, bayonet type connector.

3.4.21.4 Final check valve. - The final air discharge shall be provided with a check valve which shall be sufficiently tight to permit unloading and depressurizing of the compressor at shutdown.

3.4.21.5 Small manually operated valves such as gage cocks, separator drain valves, and similar items shall be of the soft seat type or tapered needle type with replaceable packing. Such valves in cooling water systems may be stainless steel, class 316, in accordance with QQ-S-763.

3.4.22 Air cleaner silencers. - Provisions shall be made for cleaning the inlet air and silencing the inlet air noise. These functions may be performed by a combination cleaner silencer or by a separate device for each function. Provision shall be made for replacement of cleaner elements. The inlet connection shall be flanged for possible attachment of an external pipe to conduct clean air to the compressor from a remote location. These devices shall be as compact as practicable consistent with effective sound attenuation (see 3.4.6) and filtration.

3.4.23 Relief valves. - Air relief valves shall be of the soft seat type in accordance with MIL-V-22549. A valve shall be installed on or after each intercooler and aftercooler in a location where the air temperature does not exceed 180°F. Water relief valves shall be in accordance with MIL-V-20065 except that inlet and outlet connections shall be silver-brazing unions in accordance with MIL-F-1183, and their discharges shall be piped to a common drain connection suitable for attachment to ship's piping. The manufacturer's piping shall be sized so as not to affect the operation of the relief valves.

3.4.24 Piping. -

3.4.24.1 All piping and fittings shall conform to MIL-STD-777 except as modified herein. Where ultrasonic inspection requirements are invoked, components to the following drawings shall be used; 810-1385941 to 810-1385946 inclusive, 810-1385948, 810-1385950 to 810-1385952 inclusive.

3.4.24.2 Unions or flanges shall be used for compressor water connections to external shipboard piping systems. The unions and flanges shall be in accordance with MIL-F-1183 and MIL-F-20042, respectively. When space or weight savings can be effected, terminal connections to compressor component parts may have unions partially integrated with such parts, and may utilize the brazed or welded ring and male union nut type of construction with soft copper gaskets. Tapered pipe thread connections shall not be used except for fittings such as plugs for core holes, or the dead ends of cross-drilled passages, which are intended to keep openings permanently closed. Terminals and joints of small tubing lines mounted on the compressor shall be straight thread flareless fittings conforming to MIL-F-21467 except that working pressures above 3000 psi will be permitted in the 1/4 inch outside diameter (o.d.) size only. The final discharge air connection to the ship's piping shall be a union connection in accordance with Drawing 810-1385884 with socket welding tail piece. All piping shall be designed and arranged to permit maximum accessibility to compressor parts requiring periodic inspections or repair.

3.4.24.3 All silver brazed piping connections shall conform to the fabrication and inspection requirements of NAVSHIPS 0900-001-7000.

3.4.25 Threaded parts. - All threaded parts shall conform to Handbook H-28. Through bolting should be used whenever possible; if not possible, studs are preferred to tap bolts or capscrews.

3.4.26 Welding and allied processes. - Welding and allied processes shall be in accordance with MIL-STD-278. Welding of cast nodular iron parts is not acceptable.

3.4.27 Jacking device. - Suitable provision shall be made for jacking the compressor over if it is impractical to turn the machine by hand.

3.4.28 Painting. - Ferrous castings and weldments shall be cleaned by an acid picking process and shot blasting (not sandblasting) to remove all scale, rust and loose weld beads. After machining, ferrous castings and weldments shall be treated by a manganese phosphate rust inhibiting process (Parkerizing or equal).

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After assembly, all surfaces to be painted shall be thoroughly cleaned and free from oil and grease. All external nonworking ferrous and aluminum surfaces shall be painted with paint equivalent to the following:

(a) One coat of pretreatment primer in accordance with MIL-P-15328.
(b) One coat of zinc-chromate primer in accordance with TT-P-645, or one coat of red lead primer in accordance with MIL-P-17545.
(c) One finish coat of light gray equipment enamel in accordance with MIL-E-15090.

Paint shall not be applied to any internal surface. When desired for convenience or appearance, contiguous non-ferrous surfaces may also be painted.

3.4.29 Identification plates.-

3.4.29.1 Identification plates shall be furnished on each compressor and shall be type A or D in accordance with MIL-P-15024, except that identification plates of plastic or aluminum are not acceptable. The plates shall be secured to equipment with corrosion resistant metallic fasteners.

3.4.29.2 Identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Classification.
(d) Manufacturer's serial number.
(e) Salient design characteristics:

(1) Capacity.
(2) Pressure in p.s.i.
(3) Speed of shaft in r.p.m.
(4) Brake hp.
(5) Working pressures of each stage.
(6) Lubricating oil pressure.
(7) Any other data vital to the unit.

(f) Contract number (and item number for multiple unit orders).
(g) Federal stock number.
(h) Section for inspector's stamp.

3.4.29.3 Each driving unit and each major accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.5 Materials.-

3.5.1 For the materials of construction of major parts of the compressors covered by this specification, see table I of 6.4. This table lists a number of materials which are considered adequate for certain critical applications. However, this specification is not intended to be restrictive provided proposed alternate materials will give equal or better service, and Bureau of Ships approval is obtained prior to offering equipment in response to invitations for bids. Manufacturers are urged to make use of class C drawings (basic design drawings) to obtain approval of any desired alternate materials (see 3.8.2.3).

3.5.1.1 Table I shall be followed in accordance with the following:

(a) Where material specifications are listed in table I, the selected specification or the approved alternate shall be used and shall be indicated on the drawings.
(b) Where material specifications are not listed for specific items in table I, the material shown or its approved alternate shall be used and shall be indicated on the drawings. The material shall be of a commercial grade or specification satisfactory for the application.
(c) Material selection for items or parts not listed in table I or for items not limited or defined by requirements contained elsewhere in this specification shall be the responsibility of the manufacturer and shall be suitable for the service required. The material selected shall be shown on the drawings.

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(d) Material specifications for items covered by 3.5.1.1 (b) and (c) shall be shown on the drawings, for information only; and the drawings shall indicate by note that the referenced specifications for these items will not necessarily be followed. Physical and chemical test records are not required.

3.5.2 All internal ferrous parts in the compressed air path shall be corrosion resistant material or shall be given a corrosion resistance treatment as approved by the Bureau of Ships. This treatment shall be completely described on the drawings. Cast gray iron shall not be used except for cylinder liners, pistons and piston rings.

3.5.3 All gaskets, "O" rings, packing, or other non-metallic parts coming in contact with lubricating oil, with compressed air having entrained oil or oil vapors, or with moisture condensate containing particulate or emulsified oil, shall be of a material compatible with petroleum lubricants.

3.6 Electrical equipment.-

3.6.1 Motors.- All motors shall be in accordance with MIL-M-17060 for a.c. motors, and the following:

(a) Service	A.
(b) Voltage and current-characteristics	As specified in the contract or order.
(c) Enclosure	Dripproof protected to 30 degrees inclination.
(d) Ambient temperature	50°C.
(e) Duty	Continuous.
(f) Speed	Constant.
(g) Winding	Squirrel cage induction.
(h) Insulation	Class A or B
(i) Bearings	Ball as specified in MIL-M-17060, or spherical double row as specified in FF-B-185.

3.6.1.1 The horsepower (hp) rating of each motor shall be in accordance with the motor ratings, but not less than 1.05 times the brake hp of the driven compressor under rated conditions of capacity and pressure. Intermediate hp ratings shall be considered for motors built integrally with the compressors. Compressor balancing rings are not required for motors built integrally with the compressor.

3.6.2 Controllers and master switches.- All controllers and master switches shall be in accordance with MIL-C-2212 for a.c. equipment, and the following:

Enclosure	Dripproof.
Operation	Magnetic.
Ambient temperature	50°C.
Insulation	Class B (class A for coils only).
Performance	(See 3.6.2.1.)
Control function	Motor starting.
Type	Across-the-line.
Protection	Overload, low voltage protection (LVP).
Duty	Same as motor (see 3.6.1).
Master switches required	(See 3.6.2.1.)
Master switch enclosures	Spraytight (remote); dripproof (local).
Master switch location	Remote or local.

3.6.2.1 Each controller shall be provided with remote "START" and "STOP" pushbuttons, a local or remote two-position switch with contacts maintained open in "MANUAL" and closed in "AUTOMATIC" position, one or more remote pressure switches and an automatic timing device. Each controller shall be equipped with a suitable non-resettable hour recording meter and events recording meter to record total cumulative operating time and cumulative total number of starts. These functions may be performed by two separate instruments located on the door of the controller cabinet.

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3.6.2.2 With the selector switch in the "MANUAL" position, the compressor shall start when the "START" pushbutton is pressed, provided the pressure switch, temperature monitor and overload relay contacts are closed. The compressor shall then continue to run in a completely loaded condition until the "STOP" pushbutton is pressed, or until the pressure switch, temperature monitor or overload relay contacts open, in which event the compressor shall stop and shall not then restart until the "START" pushbutton is again manually depressed. In this condition the pressure switch control function is to provide over-pressure protection only. With the selector switch in the "AUTOMATIC" position, pressing the "START" pushbutton shall place the compressor in operation under automatic control of the pressure switch. Provided the temperature monitor and overload relay contacts are closed when the air pressure in the discharge line is below the pressure switch cut-in pressure setting, the pressure switch contacts shall close and start the compressor. The compressor shall continue to run in the completely loaded condition until the pressure rises to the preset cut-out pressure and opens the pressure switch contacts. Opening the pressure switch contacts shall cause the unloading devices to operate and unload the compressor but shall not stop the compressor. The compressor shall then continue to operate in the unloaded condition until the pressure in the discharge line drops to the preset cut-in pressure and closes the pressure switch contacts. Closing the pressure switch contacts shall then cause the unloading devices to fully load the compressor. Each closing of the pressure switch contacts shall also cause the resetting of the automatic timing device, and each subsequent opening of the pressure switch contacts shall start the timing cycle, which at the end of any 10-minute period during which the compressor has continuously operated in the unloaded condition, shall operate to stop the compressor. In this event the compressor shall automatically restart and resume operation in the fully loaded condition only when the discharge line pressure has dropped to the cut-in point to reclose the pressure switch contacts. Under this arrangement the compressor will operate under completely automatic control but cannot continuously operate unloaded for a period in excess of 10 minutes, nor can the compressor stop and automatically restart more frequently than six times per hour.

3.6.2.2.1 Motor overload or compressor over-temperature protection shall not be bypassed by the "ON" pushbutton. Emergency run feature shall not be provided.

3.6.2.2.2 The pressure switch shall be designed so that the differential between opening pressure and closing pressure can be adjusted to any value in the range of 200 p.s.i. through 1500 p.s.i. when the closing pressure is 3000 p.s.i.

3.6.2.3 Cooling system valves and unloading valves, if electrically operated, shall conform to all the applicable requirements for controllers as specified in 3.6.2, except that enclosures shall be watertight (or better). Valves shall be tight-seated and all internal parts in contact with water shall be non-ferrous material except where ferrous parts are necessary for the magnetic circuit, and such parts shall be suitably nickel plated for protection against corrosion. Cooling system valves shall be provided with means for manually opening the valves in case of coil failures.

3.7 Repair parts and tools.-

3.7.1 Onboard repair parts.- When required (see 6.1), repair parts shall be furnished and processed in accordance with MIL-P-15137. Each set of repair parts shall consist of the parts listed on the latest approved Allowance Parts List (APL) covering the particular equipment being furnished under the contract or order. If no approved APL exists covering the particular equipment, table II shall be used as the basis for provisioning forms required by MIL-P-15137, and any other items considered necessary by either the bureau or agency concerned or the manufacturer to be essential in successful servicing of the equipment shall be included.

3.7.2 Repair parts for motors, controllers, pressure switches and other devices shall be furnished as required by the applicable specifications.

3.7.3 Special tools.-

3.7.3.1 Maintenance tools.- Each set of special maintenance tools required by a contract or order shall consist of tools such as babbitting mandrels, trams for setting piston clearances, cooler tube expanders and test rings, and gear wheel, flywheel, and coupling pulling tools required for the proper repair and maintenance of the compressors. Each set of special tools shall be furnished in a separate steel box. Note to ordering activities: Normally the number of sets of special maintenance tools should be the same as the number of sets of onboard repair parts; however, the number of sets may be increased as required by the size and organization of the ships.

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Table II - Onboard repair parts.

Parts as applicable to compressors furnished	Quantity of parts furnished per set of repair parts							
	Number of compressors having identical parts							
	1	2	3	4	5	6	7	8 or more
Bearings, complete sets	2	2	3	4	5	6	7	8
Bolts, connecting rod, nuts, washers and so forth (crank end), complete sets	1	1	2	2	3	3	4	4
Cable, temperature sensing element (length sufficient to replace three of longest leads)	1	1	1	1	1	1	1	1
Cylinder heads, sets	1	1	1	1	1	1	1	1
Cylinder liners, each size	1	2	3	4	5	5	5	5
Cylinder lubricator wearing parts, sets	1	2	3	4	5	5	5	5
Couplings, flexible, complete	1	1	1	1	1	1	1	1
Filter or strainer element, each size and type	1	1	1	1	2	2	2	2
Gages, oil level	1	1	2	2	3	3	4	4
Gages, pressure, sets	1	1	1	1	2	2	2	2
Gaskets and packing, sets	4	8	12	16	20	24	28	32
Packing, rod or piston, sets	1	1	2	2	3	3	4	4
Piston rings, sets, for each stage	4	8	12	16	20	24	28	32
Pistons, each size	1	1	1	1	1	1	1	1
Plunger, packed each size	1	1	1	1	2	2	2	2
Pumps, oil, complete 1/	1	1	1	1	1	1	1	1
Pumps, water, wearing parts for, sets	1	2	3	4	4	4	4	4
Rod, piston, packed, each size	1	1	1	1	2	2	2	2
Rods, connecting, complete sets	1	1	1	1	1	1	1	1
Springs for automatic control devices, sets	1	2	3	4	5	6	7	8
Springs, inlet and discharge valve, sets	4	8	12	16	20	24	28	32
Springs, relief valve, sets	1	2	3	4	5	6	7	8
Temperature device parts as follows:								
(a) Transformers, each type and size	1	1	1	1	1	1	1	1
(b) Rectifiers, each type and size	2	2	2	2	2	2	2	2
(c) Test cable assembly	1	1	1	1	1	1	1	1
(d) Relays, each size and type	1	2	3	4	5	6	7	8
(e) Resistors, each size and type	1	2	3	4	5	6	7	8
(f) Potentiometer, each size and type	1	1	1	1	1	1	1	1
(g) Indicating meter	1	1	1	1	1	1	1	1
(h) Selector knob	1	1	1	1	1	1	1	1
(i) Fuses, each size and type	10	20	30	40	50	50	50	50
(j) Light bulb	2	4	6	8	10	12	14	16
Temperature sensing elements, each size and type	1	1	1	1	2	2	2	2
Tube nests complete, intercooler and aftercooler, sets 2/	1	1	2	2	3	3	4	4
Tubes, intercooler and aftercooler, where easily renewable, percent of each size and type	10	10	20	20	30	30	30	30
Valves, discharge, for all stages, complete with seats, guards, springs and so forth, sets	1	1	2	2	3	3	4	4
Valves, inlet for all stages, complete with seats, guards, springs, and so forth, sets	1	1	2	2	3	3	4	4
Valve discs or strips, suction and discharge, complete sets	2	3	4	5	6	6	6	6
Working parts for automatic control devices, sets	1	1	2	2	3	3	4	4
Wrist pins, sets	1	1	2	2	3	3	4	4
Zinc protectors, sets	2	4	6	8	10	12	14	16

1/ Pump wearing parts only shall be furnished when pumps are built integrally with the compressor so that complete assembly cannot be furnished.

2/ Complete tube nests are required only when tubes are welded or brazed.

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3.7.3.2 Operating tools.- Each set of special operating tools required by a contract or order shall consist of tools, such as valve wrenching plates, normally required to operate one compressor. Each set of special operating tools shall be furnished in a separate steel box. Note to ordering activities: Normally a set of special operating tools shall be ordered for each compartment in which an air compressor of the type ordered is installed.

3.7.3.3 Tools which are available in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Government inspector), such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed and included in the list of material.

3.7.4 Stock.- The selection, stock numbering and quantities of repair parts for stock shall be in accordance with MIL-P-15137.

3.7.4.1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania.

3.8 Drawings.- All drawings and lists shall conform to MIL-D-963, with exceptions and additional requirements contained herein.

3.8.1 Drawings of the following classes shall be furnished as required by the contract or order:

Class A - Masterdrawings.
Class B - Certification data, outline drawing, drawing lists.
Class C - Basic design drawings.
Class D - Ship equipment drawings.

3.8.2 Class A and class D drawings.- Class A and class D drawings shall be so complete as to permit:

(a) Evaluation of performance and quality of the equipment against requirements of this specification.
(b) Evaluation of operational and personnel safety.
(c) Procurement of parts by the supply system.
(d) Installation, operation, maintenance and repair, without manufacturer's assistance.

3.8.2.1 Class A and class D drawings, covering air compressors under this specification, are identical as far as form and content are concerned. Class A drawings and any modifications thereof shall be approved only by the Bureau of Ships. Bureau of Ships field representatives may approve the drawings as class D drawings only (thereby authorizing manufacture without waiting for Bureau of Ships action), and at the same time or subsequently may recommend to the Bureau of Ships that the drawings be considered as class A drawings.

3.8.2.1.1 Master plans or type I drawings furnished under earlier editions of MIL-D-963 are considered as fulfilling the requirements of class A drawings under this specification regardless of the source of approval of those drawings. However, approval of modifications to those drawings after the date of this specification shall be by the Bureau of Ships only.

3.8.2.2 Class B drawings.- Class B drawings shall be furnished as a supplement to all class A and class D drawings required under a contract or order. They are not required in connection with class C drawings.

3.8.2.3 Class C drawings.- Class C basic design drawings covering air compressors under this specification may be submitted to the Bureau of Ships prior to and independent of invitation for bids or requests for proposals or quotations.

3.8.2.3.1 Class C drawings shall show basic design with sufficient clarity, completeness and detail as to permit a thorough evaluation of characteristics, performance and quality of the design against the requirements of this specification and to provide reasonable assurance to the Bureau of Ships that air compressors manufactured in accordance with the design will meet the requirements of this specification.

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3.8.2.3.2 A class C drawing may indicate a variation in materials from that called for in table I, and if approved by the Bureau of Ships that variation in material shall take precedence over the corresponding material in table I subject to 3.8.7.1. All such material variations shall be specifically pointed out in the transmittal letter. Approval of a class C drawing does not preclude the requirement for class A or D drawings and class B drawings applicable to a particular contract or order.

3.8.3 Format.- The manufacturer's commercial format may be used if the requirements for drawing content (see 3.8.4), material identification (see 3.8.5), and drawing identification (see 3.8.6) are satisfied.

3.8.4 Drawing content.-

3.8.4.1 General requirements for all classes.-

3.8.4.1.1 "Manufacturer's Use Only" notes.- Information intended for manufacturer's use only shall be designated. Conversely, on a commercial drawing with wide usage, it is permissible to designate portions thereon "For Navy Use Only".

3.8.4.1.2 Dimensional tolerance.- Unless otherwise indicated by note, legend, or key, it will be assumed that all tolerances shown on a drawing must be complied with during both manufacture and maintenance repair.

3.8.4.2 Type and contents of class A and class D drawings.- Drawings of the types specified in 3.8.4.2.1 through 3.8.4.2.3.1 shall be furnished.

3.8.4.2.1 Sectional assembly drawings.- Sectional assembly drawings shall include a sectional assembly and elevation drawings with complete list of material, with references to applicable detail drawings. The list of material shall include every part required in the compressor assembly, including those parts not required to be detailed. This may necessitate some side or partial views in order to show parts not otherwise shown in the main section view. The list of material shall include an indication of each part required by this specification to be furnished as an onboard repair part (this is not a repair parts list).

3.8.4.2.1.1 If more than one sectional assembly or elevation drawing is required to identify all parts, or if the sectional assembly with list of material results in a larger drawing than size 34 by 44 inches, then the list of material may be on a separate master drawing.

3.8.4.2.1.2 Sectional assembly drawings shall show all critical assembly and running clearances. The sectional assembly drawings shall also reference the preproduction test, high-impact shock test, and inclination test reports, and the bureau or agency letters approving those reports, except that when a general arrangement drawing (see 3.8.4.3.1) is used, this data may be referenced on the general arrangement drawing.

3.8.4.2.2 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "List of Material" of the sectional assembly drawing.

3.8.4.2.3 Detail drawings.- Detail drawings shall be furnished of all parts and subassemblies necessary for evaluation of the equipment, and of all parts necessary for maintenance, and overhaul of the equipment. Details of these parts shall be so complete as to permit emergency manufacture by a Naval shipyard without assistance from the original manufacturer. Details of such parts as crankcases and cylinder blocks shall be furnished, but the dimensioned development of cored hydraulic passages need not be shown. Subassemblies whose parts can not be procured or serviced individually should be shown as a single part and so indicated. Multidetail drawings are preferred, but monodetail drawings may be used.

3.8.4.3 Contents of class B drawings.- A class B outline drawing including certification data shall be furnished as a supplemental drawing to all class A and class D drawings. Outline drawings shall cover complete units, including compressor, driver, bedplate and accessories. Separate class B drawings shall be furnished under each contract or order unless the complete equipments covered by the outline drawing and the referenced drawings are in fact identical in all respects to previously approved class B drawings.

3.8.4.3.1 If the compressors and motors are close-coupled or built together so that they are not in fact separate component parts, the outline drawing may be considered as a master drawing, entitled

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"General Arrangement and List of Reference Drawings", and all certification data applying to an individual contract or order (including electrical controller reference drawings) should be omitted from the general arrangement drawing. In this case separate certification data sheets, including all data applicable to the individual contract or order, shall be furnished, and the list of repair parts may be included in the certification data sheets or on a separate master drawing to be referenced by the certification data sheets.

3.8.4.3.2 In addition to the certification data required by MIL-D-963, the class B drawings or "General Arrangement" drawings shall contain the following:

(a) Complete performance data of compressors, prime mover, and all auxiliaries.
(b) Table of weights of individual components and weight of complete unit.
(c) Identification of piping fittings which connect to shipbuilder's piping, showing their size and type.
(d) Center of gravity of the compressor component and of the complete assembly.

3.8.4.3.3 Onboard repair parts list. - A list of onboard repair parts shall be furnished with class B drawings. This list may be prepared on "Provisioning List" forms in accordance with MIL-P-15137 with the additional requirement that the list shall be assigned a manufacturer's drawing number. The forms shall be assigned sheet 2, sheet 3 and so forth of the class B drawing. This list shall be furnished whether or not the parts are required in a compressor contract or order. The list shall not be modified to indicate parts or quantities of parts furnished under supplementary or separate contracts or orders. The list is intended as a record of recommendations of the manufacturer and compressor purchaser at the time of the compressor purchase. The list of onboard repair parts actually purchased shall be proposed as required by MIL-P-15137 and need not include a manufacturer's drawing number.

3.8.4.4 Contents of class C drawings. - The class C drawings shall consist of small scale (not necessarily to scale) plan and elevation views showing overall dimensions. The range of capacities, pressure and speeds shall be shown. These drawings shall include an undimensioned sectional assembly with complete list of material which shall include every part required in the compressor and auxiliary assemblies. Cooling systems, lubrication systems of running gear and cylinders shall be clearly delineated. Alternate subassembly arrangement of different parts may be included as desired.

3.8.4.4.1 Detail drawings of individual parts are not normally required in connection with class C drawings, but may be included if desired by the manufacturer for clarification. However, detail drawings of individual parts shall be included in specific cases if required by the Bureau of Ships for evaluation and approval.

3.8.5 Material identification. -

3.8.5.1 Preferred material reference. - Where material specifications are required by 3.5 to be indicated on the drawings, and where materials of identical or equal quality can be identified by more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced, the following is the order of preference:

(a) Federal specification.
(b) Military specification.
(c) Industry and Technical Society specification or standard.
(d) Manufacturer's specification or standard.

3.8.5.2 Material substitutions. - Where materials other than those covered by (a), (b), and (c) of 3.8.5.1 are referenced and approved, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the material in terms of the Federal or Military specification most nearly applicable in order to enable Naval repair facilities to make emergency repair as necessary from Navy material stocks.

3.8.6 Drawing identification. - As indicated in MIL-D-963, Bureau of Ships drawing numbers are no longer assigned; hence, the manufacturer's drawing number is now the sole reference identification. "Manufacturer" in this case is defined as the company which produces the part, or parts, shown on the drawing.

3.8.6.1 Prime vendors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the part(s) delineated thereon are produced

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by the subcontractor. The prime vendor shall not add his drawing number to the drawing except as an unofficial reference outside the drawing border or margin.

3.8.6.2 Prime vendors who purchase parts from subcontractors for final production, test or selection phases in their own shops have the option of using as the single drawing identification either their own title block and drawing number or the title block and drawing number of the subcontractor, but not both.

3.8.7 Prints for approval.-

3.8.7.1 Approval of drawing by either the Bureau of Ships or its field representatives does not relieve the vendor or manufacturer of complete responsibility for furnishing equipment meeting all performance requirements and characteristics required by the contract or order.

3.8.7.2 Prints of the following class A and class D drawings shall be submitted for approval prior to manufacture of equipment: Sectional assembly and elevation drawings (see 3.8.4.2.1), outline drawings and detail drawings of the following parts: Bedplates, crankshafts, crankcase sections, cylinders, pistons, connecting rods, air valves, intercoolers, aftercoolers, separators, diagrams of all fluid systems, bearings, temperature monitoring system, lubricators and gage board layout. (Note: If manufacture has been authorized immediately upon signing of a contract based on previously approved class C drawings, then class A or class D drawings shall be submitted for approval within 60 days after the date of the contract.)

3.8.7.2.1 After approval, each drawing shall be lettered or stamped with the approval notation which shall include reference to the letter of approval and the drawing class for which approval is received (class A and class C from Bureau of Ships only). In addition, the notation on the sectional assembly drawing shall indicate satisfactory compliance with this compressor specification.

3.8.7.2.2 Responsibility for all other detail drawings (3.8.4.2.3) rests with the vendor. Each such drawing shall bear a notation indicating review and approval by a responsible official of either the vendor or his subcontractor. The notation shall include the name or initials of approving official, his title and the date of approval. Prints of these drawings need not be furnished to the Bureau of Ships or its field representatives prior to delivery of the final drawings, unless specifically requested.

3.8.7.3 Prints of class B drawings shall be submitted for approval at the same time as submittal of the class A or class D drawings. If the class B drawing is submitted covering previously approved class A drawings, prints of the class A drawings shall accompany the class B drawing for information.

3.8.7.4 Prints of class C drawings shall be submitted to the Bureau of Ships for approval (see 3.8.2.3).

3.8.8 Drawings for drivers and associated equipment shall be in accordance with the specifications for those equipments.

3.9 Equipment manuals (technical manuals) (instruction books).-

3.9.1 Unless otherwise specified in the contract or order, equipment manuals shall be in accordance with requirements for type I manuals of MIL-M-15071. A separate manual shall be furnished for each different compressor unit except in special cases when manuals covering more than one compressor may be approved. Preliminary manuals submitted for approval shall include all proposed sections complete.

3.9.2 Illustrations.- The following illustrations shall be included in the manuals:

(a) A minimum of two photographs of the compressor, taken 180 degrees apart on a horizontal plane.
(b) Reduced size reproductions of approved outline, assembly and sectional drawings of the complete unit.
(c) Approved drawing excerpts and exploded views of parts or assemblies requiring detailed instruction.
(d) Master drawings covering the motors and controllers.
(e) Complete operation and maintenance instructions covering electrical equipment.

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3.9.3 Distribution. - Two copies of manuals shall be packed with each compressor unit. Additional copies of manuals shall be furnished as specified (see 6.1). If final manuals are not available at the time compressors are shipped, two copies of preliminary manuals shall be packed with each compressor unit; and the final manuals shall be delivered at a later time in accordance with MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality control system. - The supplier shall provide and maintain a quality control system acceptable to the Government. The supplier is in no way relieved of the final responsibility to furnish compressors meeting the requirements of this specification.

4.1.2 Description of procedures. - Procedures shall be assembled in manual form and shall indicate organization and responsibility for the control of quality.

4.1.3 Inspection during manufacture. - The supplier shall establish and maintain inspection at appropriately located points in the manufacturing process beginning with the receipt of raw materials and subcontracted parts through final testing and preparation for shipment of completed units, to assure continuous control of quality of parts, components and assemblies.

4.1.4 The supplier shall provide and maintain gages, instruments and other measuring and testing equipment and devices necessary to assure that supplies conform to contract requirements. These devices shall be calibrated against measurement standards or designated measuring equipment at established periods to assure continued accuracy. The supplier shall prepare and maintain a written schedule for maintenance and calibration of such equipment based on the type, purpose and degree of usage.

4.1.4.1 All instruments used in preproduction and performance tests shall be calibrated before and after tests.

4.2 Inspection of material. -

4.2.1 Materials used in the compressors and accessories and identified on approved drawings by Federal and Military specifications listed in table I or elsewhere in this specification shall be subjected to inspection requirements of the applicable specification by which identified.

4.2.2 Materials used in the compressors and accessories which are identified on approved drawings by Industry and Technical Society specifications or standards shall be subjected to inspection requirements of the applicable specification or standard by which identified.

4.2.3 The supplier shall establish a system of inspection and identification to insure that specified and approved materials are used in accordance with approved drawings.

4.2.3.1 Upon request of the Government, the supplier shall furnish samples of materials and available information concerning their quality and use. When the identity or quality of an item is in doubt, and in the absence of valid and acceptable test data, the supplier shall conduct such tests as are necessary to determine or verify its identity or quality.

4.3 Preproduction inspection. - Preproduction inspection shall consist of the examination and tests specified in 4.3.1 through 4.3.2.5.1.

4.3.1 Prior to the offering of any identical air compressor for acceptance by the Government, an air compressor shall have successfully undergone preproduction tests as described hereinafter, except as specified in 4.3.1.1 through 4.3.2.5.1.

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4.3.1.1 A successful preproduction test performed under one contract will not have to be repeated under a subsequent contract unless a deficiency in design and workmanship develops with a compressor in service, or unless a modification in design is considered by the Bureau of Ships to be extensive enough to warrant retest.

4.3.2 The preproduction tests shall include initial performance tests required by 4.4 followed by a running test of not less than 500 hours of actual running time. The 500 hours running time shall not be continuous, but shall include at least 6 rest periods of not less than 8 hours each.

4.3.2.1 All data required by 4.4.3 shall be taken and recorded at the beginning and end of each running period, and at intervals of 4 hours or less during the running times.

4.3.2.2 Following completion of the tests specified in 4.3.2, the machine shall be disassembled sufficiently to permit examination as specified in 4.4.3.5. Changes in appearance or physical condition of parts from the original condition shall be noted and described in the test report. Note and comment on the accumulation of carbon deposits on air valves and in coolers.

4.3.2.3 If any serious wear or other deficiencies are noted, which would necessitate replacement of parts or overhaul of the compressor in less than 1000 hours of operation, these facts shall be immediately communicated to the Inspector of Naval Material and the Bureau of Ships. The compressor shall be left unassembled pending a decision of the Bureau of Ships.

4.3.2.4 If no serious wear or other deficiencies are noted, these facts shall also be immediately communicated to the Inspector of Naval Material and the Bureau of Ships for information. The compressor shall be cleaned up, worn parts shall be replaced, and the compressor prepared for shipment in accordance with requirements of the contract.

4.3.2.5 Within 30 days after successful completion of the preproduction tests, copies of a complete test report including all recorded data shall be distributed as follows:

(a) Two to the Bureau of Ships (attention Code 649).
(b) Two to the purchasing activity (if other than the Bureau of Ships).
(c) One to the primary inspection officer.
(d) One to the branch inspection office (if any).

4.3.2.5.1 Bureau of Ships approval of the test report shall be obtained prior to delivery of any air compressor. A period of 30 days, after receipt in the Bureau of Ships, shall be allowed for Bureau of Ships action.

4.4 Tests of production compressors. -

4.4.1 Hydrostatic and silver brazed joint tests. -

4.4.1.1 Hydrostatic. - The various parts of each compressor shall be hydrostatically tested in accordance with the following:

(a) Cylinders, 1-1/2 times the maximum designed working pressure.
(b) Water jackets, 265 p.s.i.g.
(c) Intercoolers and aftercoolers, 265 p.s.i.g. on the water side and 1-1/2 times the maximum design working pressure on the air side.
(d) Piping and other parts not mentioned herein and subjected to pressure shall be tested to 1-1/2 times the maximum designed working pressure.

4.4.1.2 Silver brazed joints. - All silver brazed joints in oil, air and salt water piping shall be tested as required by NAVSHIPS 0900-001-7000.

4.4.2 Capacity test methods. -

4.4.2.1 The preferred method for determining compressor capacity is by the high pressure flask method; by noting the time required to fill a flask of known volume from zero gage pressure to the rated discharge pressure of the compressor, corrected to rated speed, receiver temperature, zero humidity and

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14.7 p.s.i.a. barometer. During the time interval required to fill the flasks, the compressor discharge pressure shall be maintained constant at the rated discharge pressure. The calibration of the flask volume shall be made at the final flask temperature of 122°F. and shall be made in the presence of the Government inspector.

4.4.2.1.1 At the option of the compressor supplier or testing laboratory, the compressor capacity tests may be made in accordance with the ASME Power Test Code (PTC9-1954) as outlined in the third edition of the Compressed Air Handbook.

4.4.3 Test procedures.-

4.4.3.1 Electrical.- Motors and controllers shall be tested as required by the applicable specifications.

4.4.3.2 Sampling.- Completely assembled compressors, including driving units, temperature controls, unloading valves, automatic separator drain equipment and all other equipment and fittings shall be selected for testing in accordance with table III and tested as specified in 4.4.3.3.

Table III - Sampling for performance tests.

Number of identical units on simultaneous orders	Random selection
Up to 20	1 from every 5 units
21 to 100	1 from every 10 units, but not less than 4
Above 100	1 from every 20 units, but not less than 10

4.4.3.3 Performance tests.- Compressor units selected in accordance with table III shall be submitted to a continuous running test of 8 hours duration at normal capacity and pressure. Test is to be run under normal electrical conditions specified in the contract or order. Power consumption tests shall be conducted as a part of the tests. For these tests, the data specified in 4.4.3.4 shall be recorded at the start and finish of the run and at regular intervals, not exceeding 1/2 hour throughout the run.

4.4.3.4 Recorded data.-

(a) Mechanical:

(1) Ambient shop conditions:

a. Dry bulb ambient temperature.
b. Wet bulb ambient temperature.
c. Intake temperature (in air stream close to intake).
d. Barometer readings.

(2) Air discharge pressure from each stage of compression.
(3) Final air temperature at discharge from aftercooler.
(4) Air temperatures before and after each stage of compression.
(5) Circulating water inlet and outlet temperatures.
(6) Quantity of cooling water in gallons per minute.
(7) Revolutions per minute of driving unit and of compressor.
(8) Temperature of lube oil in crankcase.
(9) Time required to fill flask of measured volume.

(b) Electrical:

(1) Volts, amperes and frequency.
(2) Power input to motor.

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(c) Operational:

(1) Operation of the automatic temperature control.
(2) Operation of the automatic unloading devices.
(3) Operation of moisture separators. (Include record of quantity of oil and water drained manually at 15-minute intervals from each separator for a period of 2 hours during preproduction tests.)
(4) Total amount of cylinder lube oil used during test period.

4.4.3.4.1 Computed and recorded data. - The following data shall be computed from the recorded data and shall be included on each test data record:

(a) Capacity in cubic feet of air per hour actually delivered at rated discharge pressure.
(b) Brake horsepower at the compressor shaft (computed from previous test of motor).
(c) Volumetric efficiency.
(d) Overall efficiency of compressor; that is, compression efficiency versus mechanical efficiency. Theoretical air horsepower upon which efficiency is based may be calculated from either the adiabatic or isothermal based, but test data should indicate which is used.

4.4.3.4.2 During each operation test the functioning of the electrical and mechanical equipment shall be smooth and free from excessive heating, noise, and vibration; and the performance in all respects shall meet the requirements of this specification.

4.4.3.5 Upon completion of the performance tests, each compressor and its accessories shall be disassembled sufficiently to permit examination of all parts subject to wear or damage by running, these parts to include bearings, wrist pins, pistons and piston rings, cylinders and air valves. To be acceptable, no parts of the compressor shall show any signs of failure or excessive wear. The condition of the compressors at the conclusion of all tests shall be noted on the official test sheets.

4.4.3.6 Each compressor unit not selected in accordance with table III shall be subjected to tests exactly as required by 4.4.3.3 except:

(a) The duration of the running test need not exceed 4 hours, and data is required to be taken only at the start and finish of the run and once in between.
(b) The unit need not be disassembled upon completion of the performance tests, unless excessive heating, noise, vibration or erratic performance is noted during the tests.

4.4.3.6.1 If excessive heating, noise, vibration, or erratic performance is noted and is corrected, the 4-hour test shall be repeated. If the first correction does not completely eliminate the trouble, then after further correction the compressor shall be subjected to a complete 8-hour test before being offered for acceptance.

4.4.3.6.2 All recorded data and computations required for the 8-hour test are also required for the 4-hour test.

4.4.3.7 Inclined operation. - One compressor unit of each design, including the driving motors, shall be selected for tests to determine satisfactory compliance with 3.4.1.2(b). Inclined operation tests shall be without resilient mounts.

4.4.3.7.1 Only one compressor of each design shall be subjected to the inclination tests, and the tests need not be repeated on later contracts or orders unless the Bureau of Ships considers that sufficient changes have been made from the tested design to warrant retesting. Neither changes in material only of compressor construction nor changes in the driving motor shall be considered as reasons for retesting.

4.4.3.7.2 The compressor unit shall be operated at rated load for not less than 30 minutes with the base tilted 15 degrees in each of the four cardinal positions, right, left, front, and rear. During the progress of these tests, there shall be no excessive heating of any parts, and noise and vibration shall not exceed that of normal horizontal operation.

4.4.3.8 The suitability of the compressor unit for operation when a ship is rolling or pitching shall be determined by examination of the design.

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4. 4. 4 High-impact shock test. - One compressor unit, including driving motor, of each design, shall be shock tested in accordance with MIL-S-901 for class III equipment. Shock tests shall be without resilient mounts (see 3. 4. 5. 1).

4. 4. 4. 1 Only one compressor of each design shall be subjected to the shock tests, and the tests need not be repeated on later contracts or orders unless the Bureau of Ships considers that sufficient changes have been made from the tested design to warrant retesting. Changes in materials of construction may require retesting. Changes in the driving motor only shall not be considered a reason for retesting the compressor; however, changes in the motor will require testing of the motor.

4. 4. 4. 2 Consideration will be given to waiving the shock test requirement on a compressor unit provided that compressor and motor are of the same basic design as that of another compressor which has successfully passed the shock test. The decision as to whether a shock test will be waived shall rest with the Bureau of Ships.

4. 4. 4. 3 If a supplier is unable to conduct the shock tests at his own plant, he may arrange to have them conducted at a commercial or Government laboratory suitably equipped to conduct the tests. When the shock tests are to be conducted at a Government laboratory, a complete set of master and outline drawings shall accompany the equipment.

4. 4. 4. 4 When the shock tests are on a medium-weight shock machine, six blows shall be applied as described in MIL-S-901, with the mounting platform in a horizontal position and the blows applied under the approximate center-of-gravity of the assembled compressor unit. The equipment shall be tested running and at standstill on alternate blows. In addition to the six blows required above, two additional blows shall be applied with the base of the compressor unit inclined at 30 degrees from the horizontal and the blows applied under the approximate center-of-gravity. The compressor unit shall be mounted in such a way that the horizontal component of shock is transmitted across the axis in which the equipment is weakest. The blows of the inclined test shall correspond to groups II and III in table I of MIL-S-901, and the compressors shall be running during both blows. Compressors need not be compressing air during shock tests.

4. 4. 4. 5 Examination after shock test. -

4. 4. 4. 5. 1 Motor. - Examination of the motor shall be in accordance with MIL-M-17060.

4. 4. 4. 5. 2 Compressor. - Examination of the compressor shall be conducted as follows:

(a) Examine the unit carefully and note the extent of any external damage.
(b) Jack the compressor over several times by hand to insure that all working parts are free and ready for starting.
(c) Run the compressor unloaded for a short time and note any mechanical peculiarities.
(d) Gradually build up the load and carefully check any undue noises or excessive vibration or other nonconformances.
(e) Secure the compressor and disassemble for complete examination. Observe and record all effects of the shock test on the various compressor components.

4. 4. 4. 6 Failure to pass shock test. -

4. 4. 4. 6. 1 Motor. - The motor will be considered to have failed the shock test unless it conforms to MIL-M-17060.

4. 4. 4. 6. 2 Compressor. - The compressor will be considered to have failed the shock test in the event of the following:

(a) Breakage of any parts, including accessories and mounting bolts.
(b) Detachment of any part or object from the unit.
(c) Compressor requires attention or adjustment or replacement of parts to enable it to perform its principal function during and after each blow of the test.

4. 4. 4. 7 Disposition of shock-tested equipment. - Equipment which has successfully passed the shock tests will be considered acceptable for such service as designated by the Bureau of Ships. The supplier shall thoroughly recondition such units, correcting any minor imperfections or deformations, and shall

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place the equipment in satisfactory working order. The motors shall be reconditioned and tested in accordance with the requirements of the motor specifications. After reconditioning, the equipment shall be tested as specified in 4.4.3.3.

4.4.4.7.1 Equipment which has been subjected to the high-impact shock tests and has failed will not be accepted, either in whole or in part until the causes of such failure have been corrected by the supplier, and the complete equipment has been retested. However, failure of the compressor only shall not cause rejection of an otherwise acceptable motor, nor shall failure of a motor only cause rejection of an otherwise acceptable compressor.

4.4.5 Test reports.-

4.4.5.1 Test reports shall be prepared for all tests conducted by the supplier, including the shock and inclination tests. Recorded data of the compressor performance tests may be logged on supplier's forms. The following identification data for each compressor unit shall be furnished and included in the test report:

(a) Serial numbers of the compressor and driving unit.
(b) Ship for which intended.
(c) Contract or order number.
(d) Actual net weight of the unit in the "dry" and "wet" condition.

4.4.5.2 Distribution of compressor performance test reports shall be as follows:

(a) Two to the Bureau of Ships.
(b) Two to each Supervisor of Shipbuilding for each ship's worth of compressors furnished (one for supervisor's files; one for shipbuilder) for privately built ships.
(c) Two to each Naval shipyard for each ship's worth of compressors for Navy built ships.
(d) One for primary inspection officer.
(e) One for branch inspection office (if any).

4.4.5.3 Distribution of shock test reports and inclination test reports shall be as follows:

(a) Six to the bureau or agency concerned, for approval.
(b) One for the primary inspection office.
(c) One for the branch inspection office (if any).

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Compressors.-

5.1.1.1 Preservation and packaging.- Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's identification, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged level A, packed level C and marked levels A and C, respectively, in accordance with MIL-R-196, MIL-P-17286 and MIL-E-17555, as applicable.

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5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Compressors. -

5.2.1.1.1.1 Level A. - Compressors shall be preserved and packaged in accordance with the level A requirements of MIL-C-3600.

5.2.1.1.1.2 Level C. - Preservation and packaging of compressors shall be sufficient to afford adequate protection against, corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.1.2 Repair parts. - Repair parts shall be preserved and packaged in accordance with the level A requirements of MIL-R-196 and MIL-E-17555, as applicable.

5.2.1.2 Packing. -

5.2.1.2.1 Compressors. -

5.2.1.2.1.1 Level A. - Compressors shall be packed for shipment in accordance with the overseas requirements of MIL-C-3600, except that pilot pack tests are not required.

5.2.1.2.1.2 Level B. - Compressors shall be packed for shipment in accordance with the overseas requirements of MIL-C-3600, except that pilot pack tests are not required and except that shipping containers shall be of the domestic type or class.

5.2.1.2.2 Repair parts. - Repair parts shall be packed level A or B as specified in accordance with the applicable preservation and packaging specification (see 5.2.1.1.2). Unless otherwise specified in the contract or order, repair parts shall be packed separately and shipped concurrently with the basic equipment.

5.2.1.3 Equipment manuals. - Service manuals shall be packaged level A or C and packed level A, B or C as specified in the contract or order in accordance with MIL-M-15071.

5.2.1.4 Drawings. - Drawings shall be packaged level A or C and packed level A, B or C as specified in the contract or order in accordance with MIL-D-963.

5.2.1.5 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity of compressors per ship on which basis the quantity of onboard repair parts in a set shall be determined (see 3.7.1).
(c) Whether additional copies of manuals are required (see 3.9.3).
(d) Preservation, packaging, packing and marking requirements other than those required by 5.1 (see 5.2).

6.2 Prospective suppliers are urged to submit to the Bureau of Ships, Code 649, class C basic design drawings of air compressors they propose to offer under this specification for Navy shipboard use.

6.2.1 If a supplier who has an approved class C basic design drawing for air compressors covered by this specification desires to make any changes or revisions in his basic design for such air compressors,

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he must submit new class C drawings or revision(s) to his previously approved class C basic design drawings, as appropriate, for approval. In the event a change or amendment to this specification is made, the Bureau of Ships will determine whether it is of such significance to basic designs as to warrant cancelling any class C drawing approval(s) previously granted. If so, it will notify suppliers concerned in writing of such cancellation and such suppliers may, and are urged to, submit for approval new class C basic design drawings, or revisions to the formerly approved class C drawings. If no notification of cancellation is given, any basic design approval granted prior to the change or amendment is automatically extended under this specification as changed or amended. Approval of new or revised class C drawings shall be effective only with respect to contracts and orders awarded subsequent to such approvals. Conditional approval of class C basic design drawings or revisions thereto will not be made.

6.2.2 It is anticipated that the Bureau of Ships will require approximately 30 days after receipt of drawings to take approval action, but a longer time may be required if the drawings need revision or correction in order to be approved. This applies to each drawing submission, except that approval action on a relatively minor revision to a previously approved class C drawing may require less than 30 days. However, the Bureau of Ships cannot and does not undertake that it will take approval action in time for any particular procurement or purchase of air compressors under this specification.

6.3 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 *Compressor materials.* - Compressor materials are listed in table I.

Table I - Compressor materials.

Applicable	Material	Applicable document 1/ 2/ 3/
Bedplate	Cast steel	MIL-S-15083, grade B or ASTM A27, grade 60-30
	or structural steel	---
	or forged steel	MIL-S-20166
	or lead tin bronze	QQ-L-225 or ASTM A201, grade B
Cylinders	Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or nodular (ductile) iron	MIL-I-24137, class A, or ASTM A536 grade 65-45-12
Cylinder heads	Aluminum bronze	QQ-B-679 or ASTM B148
	or aluminum manganese bronze	MIL-B-16522, class 2 or ASTM B147, alloy 8B
	or valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
Cylinder liners	Nitrided steel	MIL-S-869
	or gray cast iron	QQ-I-652 or ASTM A48
	or nitralloy G	Commerical
Pistons	Cold rolled, cold drawn steel	---
	or cast aluminum	QQ-A601 or ASTM B26
	or gray cast iron	QQ-I-652 or ASTM A48
	or forged steel	---
	or stainless steel	---

See footnotes at end of table.

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Table I - Compressor materials (cont'd).

Application	Material	Applicable document 1/ 2/ 3/
Crankcases, frames and housings	Cast steel	MIL-S-15083, grade B or ASTM A27, grade 60-30
	or structural steel	---
	or nodular (ductile) iron or forged steel	MIL-I-24137, class A MIL-S-20166
	or lead tin bronze	QQ-L-225
Crankshaft	Forged steel	MIL-S-890, class An or B-S
	or nodular (ductile) iron	MIL-I-24137, class A
Flywheel	Cast steel	MIL-S-15083, grade B
Connecting rods, piston rods, caps, bolts	Forged steel	MIL-S-890, class An or B-S
Wrist pins, crosshead pins	Forged steel	MIL-S-890, Class Ac
	or nickel molybdenum	Commercial
Housing tie rods	Forged steel	MIL-S-890, class Ac or An
Condensate separators:		
1500 p.s.i. pressure and below	Stainless steel	---
	or copper nickel alloy	---
	or gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or aluminum bronze	MIL-B-16033, class 4 (4 percent nickel minimum) or ASTM B148
above 1500 p.s.i. pressure	Stainless steel	---
	or copper nickel alloy	---
	or steel	MIL-S-23009
Water pump:		
Casing and impeller	Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or nickel copper alloy	QQ-N-288
Shaft	Nickel copper alloy	QQ-N-281, class A or B or ASTM B164, class A or B
Heat exchangers (water through tubes):		
Shells	Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or gun metal nickel	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or aluminum bronze	MIL-B-23921
	or naval brass	---
	or stainless steel	---
	or copper nickel alloy	MIL-C-15726 or MIL-T-16420

See footnotes at end of table.

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Table I - Compressor materials (cont'd).

Application	Material	Applicable document 1/ 2/ 3/
Tube sheets, heads and plates	Aluminum bronze	QQ-C-450
	or aluminum bronze	QQ-B-679
	or aluminum manganese bronze	MIL-B-16522
	or copper nickel alloy	MIL-C-15726
Baffles	Copper nickel alloy or brass	--- ---
Tubes	Copper nickel alloy	MIL-T-15005 or MIL-T-16420 or ASTM B111
Heat exchangers (water surrounding tubes): Tubes	Copper nickel alloy	MIL-T-15005 or MIL-T-16420 or ASTM B111
Tube sheets, heads and plates	Copper nickel alloy	MIL-C-15726
	or valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or copper silicon alloy	QQ-C-591
	or aluminum bronze	QQ-C-450
	or aluminum bronze	QQ-B-679
	or aluminum manganese bronze	MIL-B-16522
Baffles	Copper nickel alloy or brass	--- ---
Shells	Copper nickel alloy	MIL-T-15005 or ASTM B111
	or copper nickel alloy	MIL-T-16420

1/ "ASTM" indicates standard specifications of the American Society for Testing and Materials. Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.

2/ Materials used may be in accordance with either Government specifications or ASTM specifications, or equivalent material to any other recognized industry standard specification, such as Society of Automotive Engineers (SAE) and American Iron and Steel Institute (AISI), at the supplier's option.

3/ Detail drawings and lists of material shall reference the specifications actually followed in each case and shall include the class, type or grade of material used in each case, as applicable (see 3. 5. 1 and 3. 8. 5. 1).

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6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 4310-N044SH)

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